

Electromagnetically Induced Grating via Coherently Driven Four-Level Atoms in a *N*-Type Configuration

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Abstract We propose a scheme to generate an electromagnetically induced grating via coherently driven four-level atoms in a *N*-type configuration in the presence of a standing signal field, a coupling field and a probe field. We show that a nearly ideal phase grating can be realized by adjusting the frequency detuning of signal field, the interaction length of atomic medium, and the ratio of the intensity between the signal field and the coupling field. The first-order diffraction efficiency of the grating is about 29.9 %, which is close to that of an ideal sinusoidal phase grating.

Keywords Electromagnetically induced grating · Electromagnetically induced transparency · Fraunhofer diffraction

1 Introduction

For decades, much attention has been paid to understandings and applications of electromagnetically induced transparency (EIT) [1, 2], which is a kind of quantum interference effects and arises in three-level systems. EIT is a powerful technique that can be used to make an otherwise absorbing medium transparent to a weak probe laser on resonance with a certain atomic transition by applying a strong coupling laser under certain circumstances. Recently, many applications demonstrate the importance of EIT, for example, ultraslow light [3], light storage [4, 5], quantum memory [6], greatly enhanced Kerr nonlinearities [7–9], multi-wave mixing [10], optical solitons [11–13] and electromagnetically induced grating (EIG) [14–23].

Ling et al. [14], firstly, introduced the notation of EIG, which can diffract a weak probe field into high-order diffractions, and proposed theoretical implement of EIG in three-level

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Λ -type atoms in the presence of the standing wave coupling field via EIT. Mitsunaga et al. [15] experimentally observed EIG via using a forward four-wave mixing configuration in cold sodium atoms. Cardoso et al. [16] also experimentally investigated laser diffraction on EIG in a degenerate open two-level system. Recently, Brown and Xiao [17] experimentally demonstrated all-optical switching and routing via EIG based on spatially modulated EIT in the Λ type system of rubidium vapour atoms. Dutta et al. [18] studied the formation of EIG in a Ξ -type system at the resonant and off-resonant two-photon absorption condition. Wan et al. [19, 22] investigated the effect of spontaneously generated coherence on the EIG, their results show that the diffraction efficiency of EIG is strikingly enhanced by the spontaneously generated coherence.

In this paper, we propose a scheme to generate an EIG via coherently driven four-level atoms in a N -type configuration, which is essentially different from Ling's model. In the model with three-level Λ -type atoms in the presence of the standing wave coupling field [14], the absorption and dispersion coefficients of the probe field depend on the strength of the coupling field, the EIG is implemented via changing the strength of the coupling field periodically in space. While, in our model with four-level atoms in a N -type configuration, the absorption and dispersion properties of the probe field depend on the effective two-photon detuning, which results from the ac-Stark shift via the signal field, so we could realize the EIG via adjusting the two-photon detuning periodically through changing the strength of the signal field periodically in space.

2 Physics Model

The system we consider, as shown in Fig. 1, consists of an N -type four-level atomic medium, a weak probe field with frequency ω_p , a strong coupling field of frequency ω_c , two strong signal fields of frequency ω_d (in their overlap region, a standing wave is formed in the medium). The probe field drives the $1 \leftrightarrow 2$ atomic transition with Rabi frequency Ω_p , the coupling field drives the $2 \leftrightarrow 3$ atomic transition with Rabi frequency Ω_c , while the levels $|3\rangle$ and $|4\rangle$ are connected by the standing field with Rabi frequency $\Omega_d \sin(\pi x/\Lambda)$, where Λ represents the space period and can be made arbitrarily larger than the wavelength of the signal fields by varying the angle between the two wave vectors of the corresponding signal fields.

The Hamiltonian of the total system is

$$H = \hbar \sum_{j=1}^4 \omega_j \tilde{\sigma}_{jj} - \hbar \left(\Omega_p e^{-i\omega_p t + ik_p z} \tilde{\sigma}_{21} + \Omega_c e^{-i\omega_c t + ik_c z} \tilde{\sigma}_{23} + \Omega_d \sin(\pi x/\Lambda) e^{-i\omega_d t + ik_d z} \tilde{\sigma}_{43} + c.c. \right), \quad (1)$$

where $\tilde{\sigma}_{ij} = |i\rangle\langle j|$, k_p , k_c and k_d are wave vectors, ω_j , ($j = 1, 2, 3, 4$) are the Bohr frequencies of atomic states $|j\rangle$.

We choose $H_0 = \hbar \sum_{j=1}^4 \omega_j \tilde{\sigma}_{jj} + \hbar \Delta_p \tilde{\sigma}_{11} + \hbar \Delta_c \tilde{\sigma}_{33} + \hbar (\Delta_c - \Delta_d) \tilde{\sigma}_{44}$ and consider $\sigma_{21} = \tilde{\sigma}_{21} e^{ik_p z}$, $\sigma_{23} = \tilde{\sigma}_{23} e^{ik_c z}$, $\sigma_{43} = \tilde{\sigma}_{43} e^{ik_d z}$ then in the interaction picture with respect to H_0 , the associated Hamiltonian reads

$$H_I = -\hbar \Delta_p \sigma_{11} - \hbar \Delta_c \sigma_{33} - \hbar (\Delta_c - \Delta_d) \sigma_{44} - \hbar \left(\Omega_p \sigma_{21} + \Omega_c \sigma_{23} + \Omega_d \sin(\pi x/\Lambda) \sigma_{43} + c.c. \right), \quad (2)$$

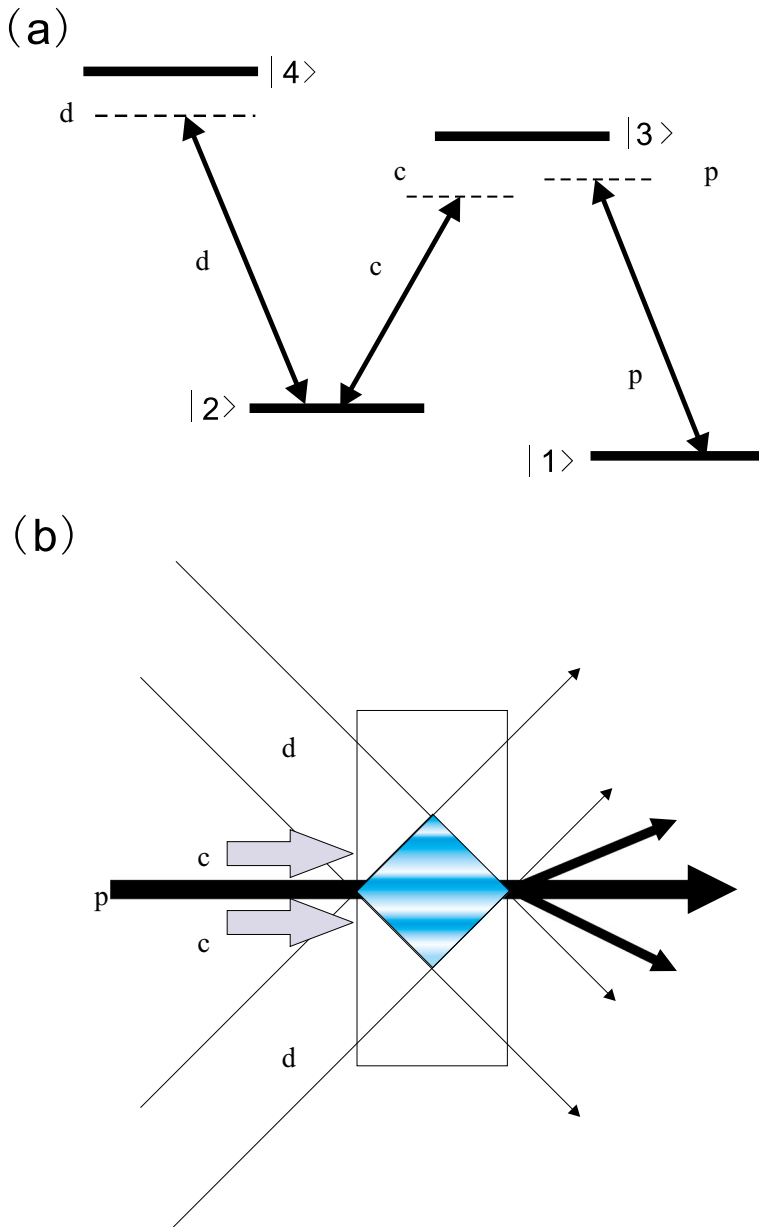


Fig. 1 **a** The energy diagram of four-level *N*-type atoms. **b** A sketch of a prototype experimental setup

where $\Delta_p = \omega_2 - \omega_1 - \omega_p$, $\Delta_c = \omega_2 - \omega_3 - \omega_c$ and $\Delta_d = \omega_4 - \omega_3 - \omega_d$ are the frequency detunings of probe field, coupling field and signal fields, respectively.

The response of the medium to the fields is governed by the density-matrix equation. The equations of motion for the density matrix elements of the four level system can be written

as follows:

$$\dot{\rho}_{21} = (-\gamma_{21} - i\Delta_p) \rho_{21} - i\Omega_p^* (\rho_{22} - \rho_{11}) + i\Omega_c^* \rho_{31}, \quad (3a)$$

$$\dot{\rho}_{31} = [-\gamma_{31} - i(\Delta_p - \Delta_c)] \rho_{31} - i\Omega_p^* \rho_{32} + i\Omega_c \rho_{21} + i\Omega_d \sin(\pi x/\Lambda_{dx}) \rho_{41}, \quad (3b)$$

$$\dot{\rho}_{41} = [-\gamma_{41} - i(\Delta_p - \Delta_c + \Delta_d)] \rho_{41} - i\Omega_p^* \rho_{42} + i\Omega_d^* \sin(\pi x/\Lambda) \rho_{31}, \quad (3c)$$

$$\dot{\rho}_{32} = (-\gamma_{32} + i\Delta_c) \rho_{32} - i\Omega_p \rho_{31} - i\Omega_c (\rho_{33} - \rho_{22}) + i\Omega_d \sin(\pi x/\Lambda) \rho_{42}, \quad (3d)$$

$$\dot{\rho}_{42} = [-\gamma_{42} + i(\Delta_c - \Delta_d)] \rho_{42} - i\Omega_p \rho_{41} - i\Omega_c \rho_{43} + i\Omega_d^* \sin(\pi x/\Lambda_{dx}) \rho_{32}, \quad (3e)$$

$$\dot{\rho}_{43} = [-\gamma_{43} - i\Delta_d] \rho_{43} - i\Omega_c^* \rho_{42} - i\Omega_d^* \sin(\pi x/\Lambda) (\rho_{44} - \rho_{33}), \quad (3f)$$

where we have introduced the dephasing rate γ_{ij} between levels $|i\rangle$ and $|j\rangle$ phenomenologically.

In the low-density approximation, the perturbation approach can be applied to the atomic part, which is introduced in terms of perturbation expansion,

$$\rho_{ij} = \rho_{ij}^{(0)} + \lambda \rho_{ij}^{(1)} + \dots, \quad (4)$$

where $\rho_{ij}^{(0)}$ is of the zeroth order in Ω_p , $\rho_{ij}^{(1)}$ is of the first order in Ω_p , and so on. By substituting (4) into (3a)–(3f) and keeping the terms up to the first order in the probe field, the equations for the first order read

$$\dot{\rho}_{21}^{(1)} = (-\gamma_{21} - i\Delta_p) \rho_{21}^{(1)} - i\Omega_p^* (\rho_{22}^{(0)} - \rho_{11}^{(0)}) + i\Omega_c^* \rho_{31}^{(1)}, \quad (5a)$$

$$\dot{\rho}_{31}^{(1)} = [-\gamma_{31} - i(\Delta_p - \Delta_c)] \rho_{31}^{(1)} - i\Omega_p^* \rho_{32}^{(0)} + i\Omega_c \rho_{21}^{(1)} + i\Omega_d \sin(\pi x/\Lambda) \rho_{41}^{(1)}, \quad (5b)$$

$$\dot{\rho}_{41}^{(1)} = [-\gamma_{41} - i(\Delta_p - \Delta_c + \Delta_d)] \rho_{41}^{(1)} - i\Omega_p^* \rho_{42}^{(0)} + i\Omega_d^* \sin(\pi x/\Lambda_{dx}) \rho_{31}^{(1)}, \quad (5c)$$

$$\dot{\rho}_{32}^{(1)} = (-\gamma_{32} + i\Delta_c) \rho_{32}^{(1)} - i\lambda \Omega_p \rho_{31}^{(0)} - i\Omega_c (\rho_{33}^{(1)} - \rho_{22}^{(1)}) + i\Omega_d \sin(\pi x/\Lambda) \rho_{42}^{(1)}, \quad (5d)$$

$$\dot{\rho}_{42}^{(1)} = [-\gamma_{42} + i(\Delta_c - \Delta_d)] \rho_{42}^{(1)} - i\Omega_p \rho_{41}^{(0)} - i\Omega_c \rho_{43}^{(1)} + i\Omega_d^* \sin(\pi x/\Lambda_{dx}) \rho_{32}^{(1)}, \quad (5e)$$

$$\dot{\rho}_{43}^{(1)} = [-\gamma_{43} - i\Delta_d] \rho_{43}^{(1)} - i\Omega_c^* \rho_{42}^{(1)} - i\Omega_d^* \sin(\pi x/\Lambda) (\rho_{44}^{(1)} - \rho_{33}^{(1)}), \quad (5f)$$

where we have assumed that the atoms are initially in state $|1\rangle$, that is $\rho_{11}^{(0)} = 1$; $\rho_{22}^{(0)} = \rho_{33}^{(0)} = \rho_{44}^{(0)} = 0$. Under the adiabatic approximation that the evolution of the atomic system is much faster than the temporal change of the radiation field, the steady state solution is found

$$\rho_{21}^{(1)} = \frac{-\Omega_p^* \gamma_{31} \Delta_d + i\Omega_p [\gamma_{31} \gamma_{41} + I_d \sin^2(\pi x/\Lambda)]}{[\gamma_{31} \gamma_{21} \gamma_{41} + I_d \gamma_{21} \sin^2(\pi x/\Lambda) + I_c \gamma_{41}] + i[I_c \Delta_d + \gamma_{31} \gamma_{21} \Delta_d]}, \quad (6)$$

where $I_c = |\Omega_c|^2$, $I_d = |\Omega_d|^2$, and we have assumed $\Delta_p = \Delta_c = 0$.

So we can obtain the wave equation for the probe field

$$\frac{\partial \Omega_p}{\partial z} - \frac{i}{N_f} \frac{\partial^2 \Omega_p}{\partial^2 x} = (\chi_r + i\chi_i) \Omega_p. \quad (7)$$

In order to present the equation in a unitless form, we choose γ_{21} as the unit for all decay rates, frequency detunings, and Rabi frequencies, Λ as the unit for x , and $z_0 = 2\hbar\gamma_{21}\epsilon_0/(Nk_p d_{21}^2)$ as the unit for z , where N is the atomic density and d_{21} is the dipole matrix element between the states $|2\rangle$ and $|1\rangle$.

Since we focus on the main features of EIG, we eliminate the transverse term by working in a parameter regime where $N_F = 4\pi\Lambda^2/(\lambda_p z_0) \gg 1$, such a requirement can be realized by increasing atomic density or Λ . Under this condition, then the equation becomes

$$\frac{\partial \Omega_p}{\partial z} = (\chi_r + i\chi_i)\Omega_p, \quad (8)$$

where

$$\chi_r = \frac{-[\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda)][\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda) + I_c\gamma_{41}] - \gamma_{31}\Delta_d^2(I_c + \gamma_{31})}{[\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda) + I_c\gamma_{41}]^2 + (I_c\Delta_d + \gamma_{31}\Delta_d)^2}, \quad (9)$$

$$\chi_i = \frac{\Delta_d\{\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda)\}(I_c + \gamma_{31}) - \gamma_{31}[\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda) + I_c\gamma_{41}]}{[\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda) + I_c\gamma_{41}]^2 + (I_c\Delta_d + \gamma_{31}\Delta_d)^2}, \quad (10)$$

then we obtain the transmission function

$$T(x) = e^{\chi_r(x)L} e^{i\chi_i(x)L}. \quad (11)$$

By the Fourier transformation of $T(x)$, Fraunhofer diffraction equation can be obtained

$$I_p(\theta) = \left| \int_0^1 T(x) \exp(-i2\pi\Lambda x \sin\theta/\lambda_p) dx \right|^2 \frac{\sin^2(M\pi\Lambda x \sin\theta/\lambda_p)}{M^2 \sin^2(\pi\Lambda x \sin\theta/\lambda_p)}, \quad (12)$$

where M is the number of spatial periods of the grating, and θ is the diffraction angle with respect to the z direction. The n th-order diffraction angle is determined by the grating equation $\sin\theta = n\lambda_p/\Lambda$. Therefore, we can obtain the first-order diffraction intensity as

$$I_p(\theta_1) = \left| \int_0^1 T(x) \exp(-i2\pi x) dx \right|^2. \quad (13)$$

3 Results and Discussions

Since we are mainly interested in the first-order diffraction, in this section, we will investigate the first-order diffraction power of the EIG under various parameters.

We first consider the simplest case $\Delta_d = 0$, that is, χ_i vanishes, which implies that no phase modulation takes place. The transmission function (11) is therefore simplified to $T(x) = \exp(\chi_r L)$, where

$$\chi_r = -\frac{\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda)}{\gamma_{31}\gamma_{41} + I_d \sin^2(\pi x/\Lambda) + I_c\gamma_{41}}. \quad (14)$$

We have plotted the transmission function $T(x)$ as a function of x in Fig. 2. From Fig. 2, we can see an obvious periodic amplitude modulation across the beam profile of the probe field. However, different from the Ling's model, at the transverse locations around the antinodes of the standing wave, the signal field intensity is very strong, then the two-photon detuning is very large, so the probe field is absorbed according to the usual Beer law. In contrast, at the transverse locations around the nodes of the standing wave, the signal field is very weak, then the two-photon detuning is very small, so the probe field is absorbed much less because of the EIT. This leads to a pure amplitude grating. To illustrate the efficiency of the first-order diffraction of this grating, we use (13) to calculate $I_p(\theta_1)$ and display the

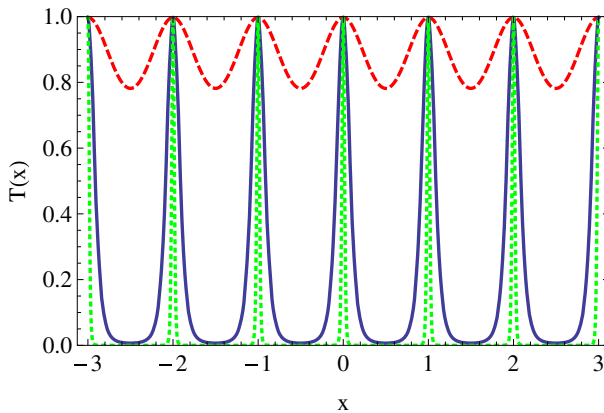


Fig. 2 The transmission function $T(x)$ as a function of x when $\Delta_d = 0$, $L = 10$, $\gamma_{41} = 0.1$, $\gamma_{31} = 0.01$, $I_c = 40$, and $I_d = 0.1$ (dashed line); $I_d = 4$ (solid line); $I_d = 100$ (dotted line)

result as a function of I_d in Fig. 3. As is well known, the diffraction efficiency of the grating depends on the grating constant, which is determined by the transmission function, so the diffraction intensity of probe field depends strongly on the intensity of the signal field I_d as shown in Fig. 3. With the increase of the intensity of the signal field, the first-order diffraction intensity gets larger quickly, and peaks at certain I_d and then fades away. This agrees with that shown in Fig. 2, with the increase of the I_d , the amplitude modulation depth increases, and at the same time, the transparency windows gets narrower.

In addition, from Fig. 3, we can see that the first-order diffraction efficiency attainable by the pure amplitude grating is limited, the maximum diffraction efficiency in our results is less than 4.5 % (the diffraction efficiency of an ideal sinusoidal amplitude grating is approximately 6.25 %). This is because that the amplitude modulation tends to gather light to the center maximum. To increase the efficiency of the first-order diffraction, we need the atomic medium, which is transparent to probe field, can also induce a π phase modulation

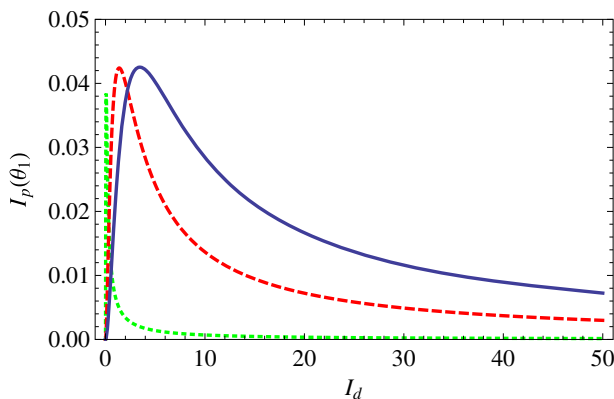


Fig. 3 The normalized diffraction intensity $I_p(\theta_1)$ as a function of I_d when $\Delta_d = 0$, $L = 10$, $\gamma_{41} = 0.1$, $\gamma_{31} = 0.01$, and other parameters are $I_c = 100$ (solid line); $I_c = 40$ (dashed line); $I_c = 2$, (dotted line)

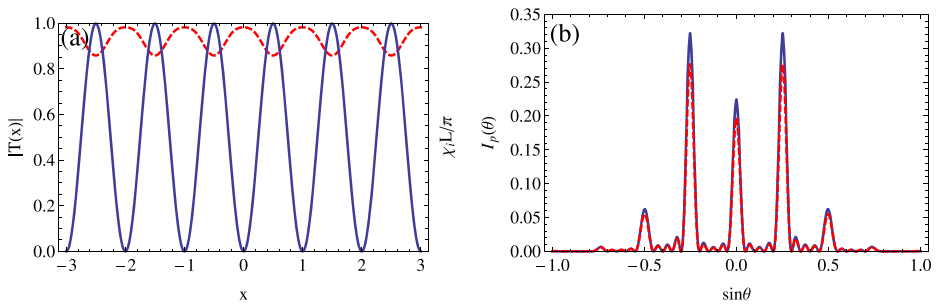


Fig. 4 **a** The amplitude of the transmission function $|T(x)|$ (dashed line) and the phase of the transmission function $\chi_i L/\pi$ (solid line) as a function of x . **b** Fraunhofer diffraction of $T(x)$ (dashed line) and pure phase modulation $\chi_i L$ (solid line) as a function of $\sin(\theta)$. The parameters are $\gamma_{31} = 0.01$, $\gamma_{41} = 0.1$, $L = 110$, $\Delta_d = 7$, $I_c = 60$, $I_d = 12$, $M = 5$ and $\Lambda/\lambda = 4$

across the probe field, since the phase modulation tends to disperse light into the high-order directions.

To induce the phase modulation, we must move into a parameter regime $\Delta_d \neq 0$. For example, when $\Delta_d = 7$, we plot the amplitude modulation $|T(x)|$ (red dashed line) and the phase modulation $\Phi/\pi = \chi_i L/\pi$ (blue solid line) for the parameters $L = 110$, $I_c = 60$, and $I_d = 12$ as shown in Fig. 4(a), which shows that we can obtain a phase modulation on the order of π with a small amplitude modulation which oscillates around $|T| = 0.93$. This indicates we can realize a grating that is very close to the ideal phase grating, which diffracts a significant portion of the probe light into the first-order patterns with $\sin\theta = \pm 0.25$ and a small portion into the second-order patterns with $\sin\theta = \pm 0.5$ as shown in Fig. 4(b).

Form the (13), we know that the first-order diffraction intensity of the phase grating depends on the interaction length L of atomic medium, the frequency detuning Δ_d of signal field. In Fig. 5, the first-order diffraction intensity $I_p(\theta_1)$ as a function of L with different Δ_d and of Δ_d with different L are plotted, respectively. Figure 5(a) shows that the first-order diffraction increases as L until it reaches a maximum value, beyond which it decreases as L with some smaller and smaller oscillations. This means there exists an optimum value of L , which is different for different Δ_d , for which the first-order intensity is significantly increased. Similarly, the first-order diffraction increases to a maximum value

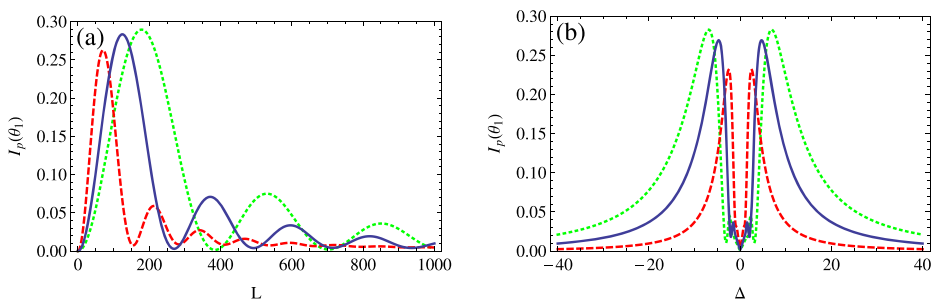


Fig. 5 The first-order diffraction intensity $I_p(\theta_1)$ as a function of L **a** with parameter $\Delta = 4$ (red dashed line), $\Delta = 7$ (blue solid line), $\Delta = 10$ (green dotted line), and of Δ **b** with parameter $L = 40$ (red dashed line), $L = 80$ (blue solid line), $L = 120$ (green dotted line). Other parameters are $\gamma_{31} = 0.01$, $\gamma_{41} = 0.1$, $I_c = 60$ and $I_d = 12$

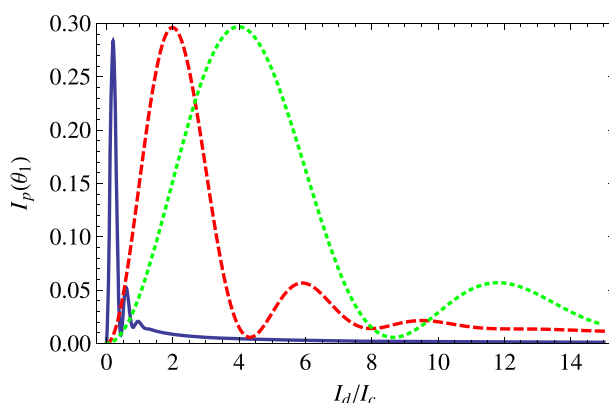


Fig. 6 The first-order diffraction intensity $I_p(\theta_1)$ as a function of the ratio I_d/I_c with parameter $\Delta = 7$ (blue solid line), $\Delta = 70$ (red dashed line), $\Delta = 140$ (green dotted line). Other parameters are $\gamma_{31} = 0.01$, $\gamma_{41} = 0.1$, $I_c = 60$, and $L = 125$

and then decreases to zero with the increase of Δ_d , there exist different optimum values of Δ_d for different L .

In addition, the first-order diffraction intensity of the phase grating also depends on the intensities of coupling field I_c and signal field I_d . In deed, if I_c is fixed, the first-order diffraction intensity depends on the ratio I_d/I_c as shown in Fig. 6. From this figure, we can see that there also exists an optimum value of the ratio I_d/I_c , which is different for different Δ_d . For example, when $\Delta = 70$, the optimum value of the ratio is $I_d/I_c \approx 1.99$, and the first-order diffraction efficiency of the phase grating is about 29.6 %; and when $\Delta = 7$, the optimum value of the ratio is $I_d/I_c \approx 0.198$, and the first-order diffraction efficiency of the phase grating is about 28.4 %. In a word, the first-order diffraction intensity of the phase grating is determined by three adjustable parameters the interaction length L of atomic medium, the frequency detuning Δ_d of signal field and the ratio I_d/I_c . Now, we can be in search of the suitable L , Δ_d and I_d/I_c to yield an optimum first-order diffraction. Through the numerical method, we find when $L \approx 179$, $\Delta_d \approx 100$, and $I_d/I_c \approx 2$, the optimum first-order diffraction efficiency of the phase grating is about 29.9 %, which is very close to that of an ideal sinusoidal phase grating about 34 %.

4 Conclusion

In conclusion, we investigated the phenomenon of electromagnetically induced grating in an atomic medium consisting of four-level N -type atoms, interacting with a standing signal field, a coupling field and a probe field. We calculated the transmission function and the intensity of the first-order Fraunhofer diffraction of the probe field when both probe and coupling fields are on resonance with their corresponding atomic transitions. We found when $\Delta_d = 0$, the EIG is a pure amplitude grating and the first-order diffraction efficiency is very limited. While a nonzero Δ_d could induced a periodic phase modulation across the probe field. We shown that we can realize a nearly ideal phase grating, the first-order diffraction efficiency of which depends on the interaction length L of atomic medium, the frequency detuning Δ_d of signal field and the ratio I_d/I_c . The optimum first-order diffraction efficiency of the phase grating is about 29.9 %, which is very close to that of an ideal sinusoidal phase grating.

Finally, it should be noted that the EIG in our model is based on the nonlinear interaction between the probe field and the signal field. However, in the model with four-level N -type atoms, the probe field is subject to EIT and the signal field is a nearly free propagating partner. The mismatch in the group velocities of two fields is a main obstacle in the four-level N -configuration scheme where the effective interaction length of the two field is reduced significantly, and the efficiency of the nonlinear interaction is strongly affected. In order to overcome the hindrance, the six-level double EIT model [24] could be used.

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